

		only 1. Using right hand grip rule, the direction of current must be out of the plane at P.
25	A	Due to the direction of current flow in X, the end of X facing Y will produce a magnetic south. As X moves away from Y, the solenoid in Y detects a reduction in magnetic south, and hence according to Lenz's Law, will generate an induced e.m.f. that may produce an effect to oppose that reduction in magnetic south, i.e. generating more magnetic north at the end of Y facing X. The nature of force will be attractive, while the induced current in Y will be flowing from N to M. The induced current will be decreasing, since according to Faraday's Law, $\frac{\Delta\phi}{\Delta t}$ is decreasing as a result of reducing flux linkage.
26	D	$B = \mu_0 n I \propto I$